WORKED SOLUTION

STEP III 2023, Question 6

Pure • Hyperbolic Functions and Monotonicity

STEP III 2023, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Both functions are a difference of $\tan^{-1} x$ and a multiple of $\tanh x$, so their derivatives are a difference of $\dfrac{1}{1+x^2}$ and a multiple of $\operatorname{sech}^2 x$. Everything therefore comes down to comparing $\cosh^2 x$ with $1+x^2$: the inequality in (i) settles the sign of f', and showing that the ratio $\cosh^2 x/(1+x^2)$ is increasing tells us exactly how often it can equal $\frac{1}{2}\pi$, which is where g' vanishes.

Solution

Part (i)

Using $\cosh^2 x = \frac{1}{2}(1+\cosh 2x)$ and $\cosh 2x = \frac{1}{2}(\mathrm{e}^{2x}+\mathrm{e}^{-2x})$, the Maclaurin series for e^x gives (the odd powers cancel)

$$\cosh 2x = \sum_{n=0}^{\infty} \frac{(2x)^{2n}}{(2n)!} = 1 + 2x^2 + \frac{2}{3}x^4 + \cdots.$$

Every term is non-negative, so $\cosh 2x \geq 1 + 2x^2$, and hence

$$\cosh^2 x = \tfrac{1}{2}(1+\cosh 2x) \geq \tfrac{1}{2}(2+2x^2) = 1+x^2,$$

with equality only when $x = 0$ (for $x \neq 0$ the x^4 term is strictly positive).

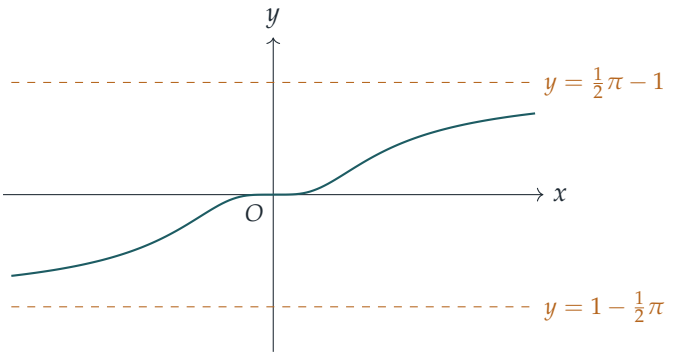
Now differentiate:

$$\mathrm{f}'(x) = \frac{1}{1+x^2} - \operatorname{sech}^2 x = \frac{1}{1+x^2} - \frac{1}{\cosh^2 x}.$$

Since $\cosh^2 x \geq 1 + x^2 > 0$, the reciprocals satisfy $\dfrac{1}{\cosh^2 x} \leq \dfrac{1}{1+x^2}$, so $\mathrm{f}'(x) \geq 0$ for all x, with equality only at $x = 0$. A function whose derivative is positive except at a single point is (strictly) increasing, as required.

Sketch. Some features to put on the graph:

- f is odd (both $\tan^{-1}$ and tanh are odd), and $f(0) = 0$.
- $f'(0) = 0$ but f is increasing, so the origin is a stationary point of inflection. In fact $\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \cdots$ and $\tanh x = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \cdots$, so $f(x) \approx \frac{1}{15}x^5$ near 0: the graph is very flat there.
- As $x \to \infty$, $\tan^{-1} x \to \frac{1}{2}\pi$ and $\tanh x \to 1$, so $f(x) \to \frac{1}{2}\pi - 1 \approx 0.571$ from below; by oddness $f(x) \to 1 - \frac{1}{2}\pi$ as $x \to -\infty$.



Part (ii)(a)

Here

$$g'(x) = \frac{1}{1+x^2} - \frac{\pi}{2\cosh^2 x},$$

which is continuous and even. At $x = 0$, $g'(0) = 1 - \frac{1}{2}\pi < 0$. At $x = 2$,

$$g'(2) = \tfrac{1}{5} - \frac{\pi}{2\cosh^2 2},$$

and $\cosh 2 > \frac{1}{2}e^2 > 3.6$, so $\cosh^2 2 > 12.9$ and $\dfrac{\pi}{2\cosh^2 2} < \dfrac{1.6}{12.9} < \frac{1}{5}$; thus $g'(2) > 0$. By the intermediate value theorem g' has a zero in $(0, 2)$, and since g' is even it has another in $(-2, 0)$. So g has at least two stationary points.

Part (ii)(b)

Let $h(x) = (1 + x^2)\sinh x - x\cosh x$. Then

$$\begin{aligned} h'(x) &= 2x\sinh x + (1+x^2)\cosh x - \cosh x - x\sinh x \\ &= x\sinh x + x^2\cosh x. \end{aligned}$$

For $x \geq 0$ both $x\sinh x$ and $x^2\cosh x$ are non-negative (and positive for $x > 0$), so h is increasing on $[0, \infty)$. Since $h(0) = 0$, we get $h(x) \geq 0$ for $x \geq 0$, and in fact $h(x) > 0$ for $x > 0$.

Part (ii)(c)

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh^2 x}{1+x^2}\right) = \frac{2\cosh x \sinh x\,(1+x^2) - 2x\cosh^2 x}{(1+x^2)^2} = \frac{2\cosh x\; h(x)}{(1+x^2)^2}.$$

For $x \geq 0$, $\cosh x > 0$ and $h(x) \geq 0$ by (b), so the derivative is non-negative; it is zero only at $x = 0$. Hence $\cosh^2 x/(1+x^2)$ is (strictly) increasing for $x \geq 0$.

Part (ii)(d)

Ask yourself How can a statement about $\cosh^2 x/(1+x^2)$ count the zeros of g'?

Multiplying $\mathrm{g}'(x) = 0$ through by the positive quantity $(1+x^2)\cosh^2 x$ shows that

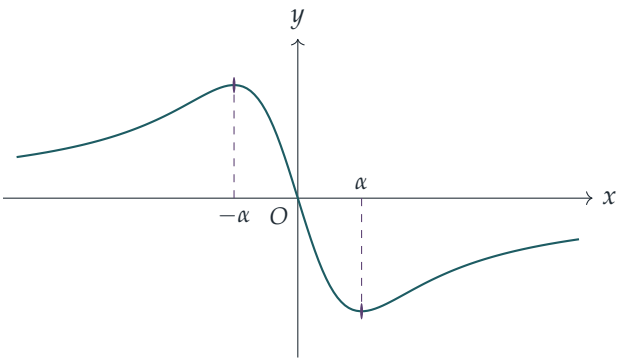
$$\mathrm{g}'(x) = 0 \iff \frac{\cosh^2 x}{1+x^2} = \frac{\pi}{2}.$$

Write $R(x) = \cosh^2 x/(1+x^2)$. On $[0,\infty)$, R is continuous and strictly increasing by (c), with $R(0) = 1 < \frac{1}{2}\pi$, so it takes the value $\frac{1}{2}\pi$ at most once there; by (a) it does take it, at some α with $0 < \alpha < 2$. So g' has exactly one zero in $[0,\infty)$. Since R is even, the zeros for $x < 0$ are the negatives of those for $x > 0$, giving exactly one more, at $-\alpha$. Hence g has exactly two stationary points, at $x = \pm\alpha$ (numerically $\alpha \approx 1.406$).

Part (ii)(e)

Collect the facts:

- g is odd, $\mathrm{g}(0) = 0$ and $\mathrm{g}'(0) = 1 - \frac{1}{2}\pi < 0$.
- From (d), $\mathrm{g}' < 0$ on $(-\alpha,\alpha)$ (where $R < \frac{1}{2}\pi$) and $\mathrm{g}' > 0$ for $|x| > \alpha$. So there is a local maximum at $x = -\alpha$ and a local minimum at $x = \alpha$, with $\mathrm{g}(\alpha) = -\mathrm{g}(-\alpha) < 0$ (about -0.44).
- As $x \to \infty$, $\mathrm{g}(x) \to \frac{1}{2}\pi - \frac{1}{2}\pi \cdot 1 = 0$. Because g is increasing on (α,∞) towards the limit 0, it stays negative there: the graph approaches the x-axis from below. By oddness, $\mathrm{g}(x) \to 0$ from above as $x \to -\infty$.
- So the only zero of g is at the origin: $\mathrm{g} > 0$ for $x < 0$ and $\mathrm{g} < 0$ for $x > 0$.



Checks

- At $x = 1$: $\cosh^2 1 \approx 2.381 \geq 2$, consistent with (i).
- $f(30) \approx 0.537$, approaching $\frac{1}{2}\pi - 1 \approx 0.571$ from below, and $f(0.1) \approx 6.6 \times 10^{-7}$, close to $\frac{1}{15}(0.1)^5$.
- Solving $\cosh^2 x = \frac{1}{2}\pi(1 + x^2)$ numerically gives $\alpha \approx 1.406$, inside $(0, 2)$ as (a) predicts, with $g(\alpha) \approx -0.440$. A fine scan of g' on $[-20, 20]$ finds exactly two sign changes, and $g(20) \approx -0.050$ is small and negative, as the sketch shows.

Common slips

- Forgetting that $\cosh^2 x \geq 1 + x^2$ must be turned round to $\operatorname{sech}^2 x \leq \dfrac{1}{1 + x^2}$; this is valid because both sides are positive.
- In (d), proving only that there is one positive root and forgetting to use evenness (or to note that $x = 0$ is not a root) to count the negative one.
- Sketching g with an asymptote other than $y = 0$, or crossing the axis again for large x; the monotonicity on (α, ∞) rules this out.

Beyond STEP

The method in (ii) (turn "how many solutions of $g' = 0$?" into "how often does a monotonic function hit a value?") is the basic way analysts count roots: continuity plus the intermediate value theorem gives existence, and strict monotonicity gives uniqueness. These ideas are made rigorous in a first analysis course; Abbott's *Understanding Analysis* is an excellent place to see them done carefully.

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